

EXPLORATORY DATA ANALYSIS USING POWER BI

REPORT WEEK - 4

Shopify Stock Analysis - DAX Measures and Visualizations

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Introduction:

The Shopify Stock Analysis project uses Power BI to analyse and visualize historical Shopify stock data. The dashboard presents stock price trends, trading volume, moving averages, and market signals using interactive charts and DAX measures. It helps users understand stock performance and identify important trends from the available data.

Objectives:

The main objectives of the Shopify Stock Analysis are:

1. To analyse historical Shopify stock data using Power BI.
2. To visualize stock price trends using Open, High, Low, and Close price data.
3. To calculate 20-Day and 50-Day Simple Moving Averages using DAX.
4. To analyse trading volume and compare it with its moving average.
5. To identify bullish and bearish trends using the Moving Average crossover signal.
6. To provide interactive date filtering using slicers for different time periods.
7. To enable dynamic metric selection so users can switch between Open, High, Low, and Close prices.
8. To provide detailed information through custom tooltips when users hover over chart data points.
9. To present complex stock data in a simple and understandable dashboard.
10. To demonstrate the use of Power BI and DAX for financial data analysis and visualization.

Steps Involved:

1. Calculate Simple Moving Average (SMA) Using DAX

A Simple Moving Average (SMA) is used to calculate the average stock price or trading volume over a specific number of days. In this project, 20-day and 50-day moving averages are calculated using DAX in Power BI.

1.1 Volume 20-Day Simple Moving Average

The 20-day volume SMA calculates the average trading volume over the latest 20 days.

Volume 20 SMA =

VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 20

VAR DateInPeriod =

DATESINPERIOD('Date Table'[Date],

CurrentDate,

-PeriodDays,

DAY)

RETURN CALCULATE(

AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),

DateInPeriod

)

The screenshot displays the Microsoft Power BI Desktop interface. The main workspace shows a DAX formula for 'Volume 20 SMA' using variables and the CALCULATE function. The formula is as follows:

```
VAR CurrentDate = MAX('Date Table'[Date])
VAR PeriodDays = 20
VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, -PeriodDays, DAY)
RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]), DateInPeriod)
```

The interface also shows the 'Measures (2)' list on the right, which includes 'Volume 20 SMA' and 'Volume 50 SMA'. The 'Properties' pane on the right shows the 'Name' of the measure as 'Volume 20 SMA' and the 'Home table' as 'Measures (2)'. The 'Data' pane on the right shows the 'Tables' and 'Model' views, with 'Measures (2)' selected. The 'Tables' pane shows the 'SHOP_2015-05-21_2025-03-16' table with columns like 'adj_close', 'close', 'date', 'high', 'low', 'open', and 'volume'.

1.2 Volume 50-Day Simple Moving Average

Volume 50 SMA =

VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 50

VAR DateInPeriod = DATESINPERIOD('Date Table'[Date],

CurrentDate,

-PeriodDays,

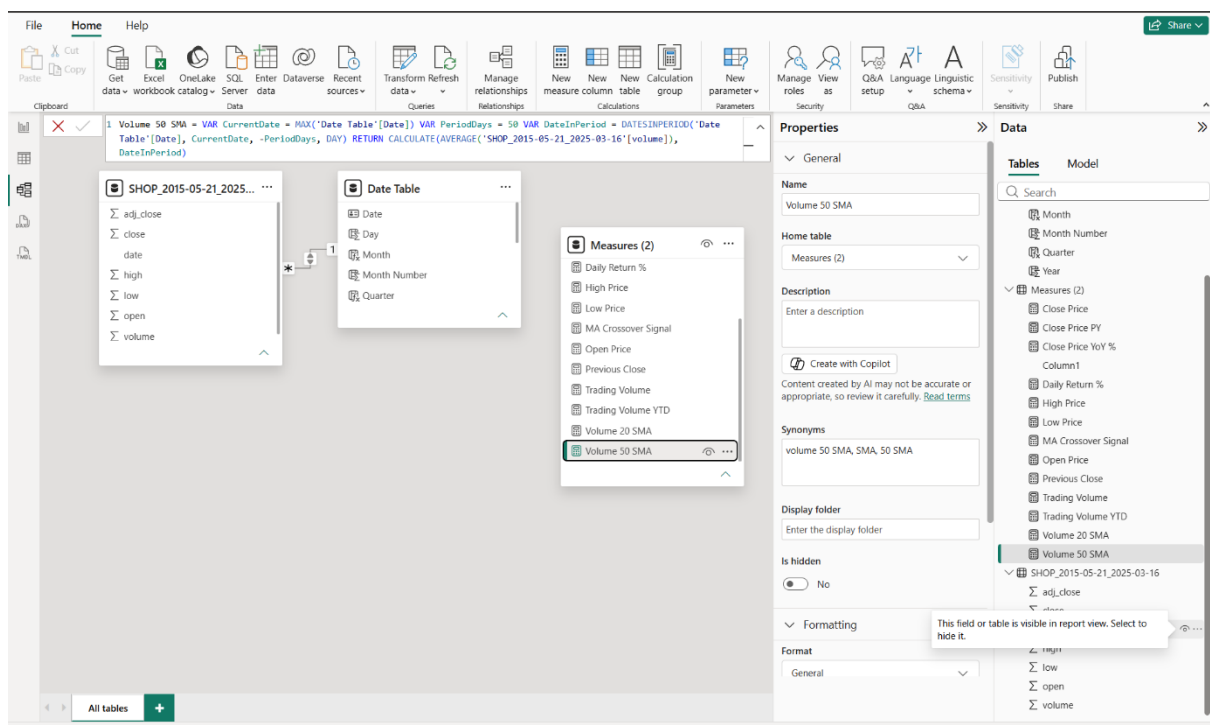
DAY)

RETURN CALCULATE(

AVERAGE('SHOP_2015-05-21_2025-03-16'[volume]),

DateInPeriod

)



1.3 Moving Average Crossover Signal

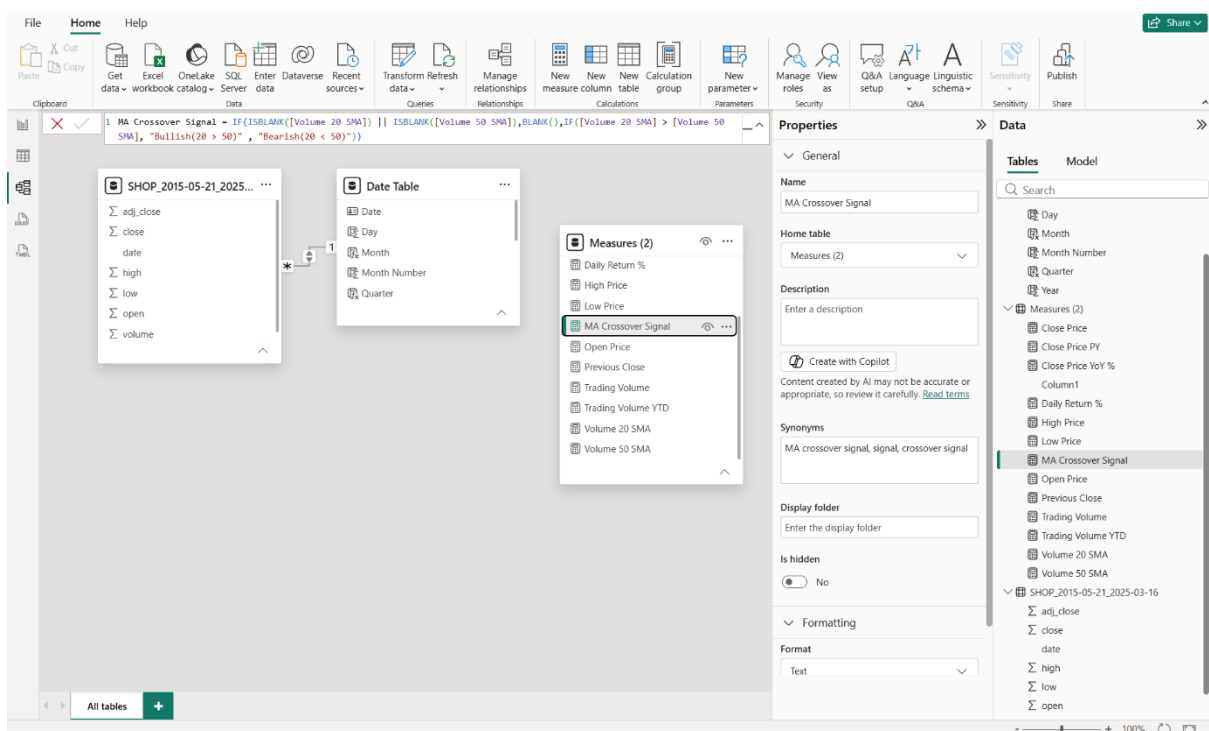
The crossover signal compares the 20-day SMA with the 50-day SMA.

```

MA Crossover Signal = IF(
ISBLANK([Volume 20 SMA]) || ISBLANK([Volume 50 SMA]),
BLANK(),IF([Volume 20 SMA] > [Volume 50 SMA],
"Bullish(20 > 50)",
"Bearish(20 < 50)"
)
)

```

This measure helps identify whether the short-term moving average is above or below the long-term moving average.



2. Creating Visualizations in Power BI

After creating the DAX measures, the next step is to create interactive visualizations.

2.1 Price Trend Chart

Select **Line and Stacked Column Chart** from the Visualizations pane.

Configure it as follows:

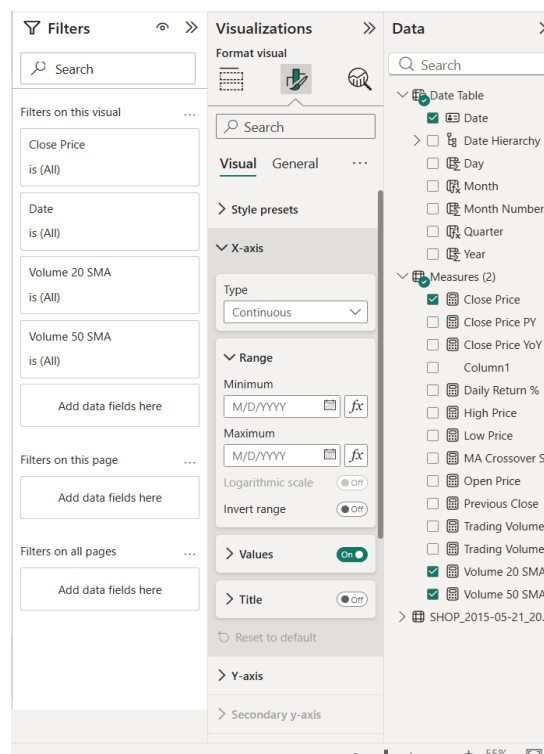
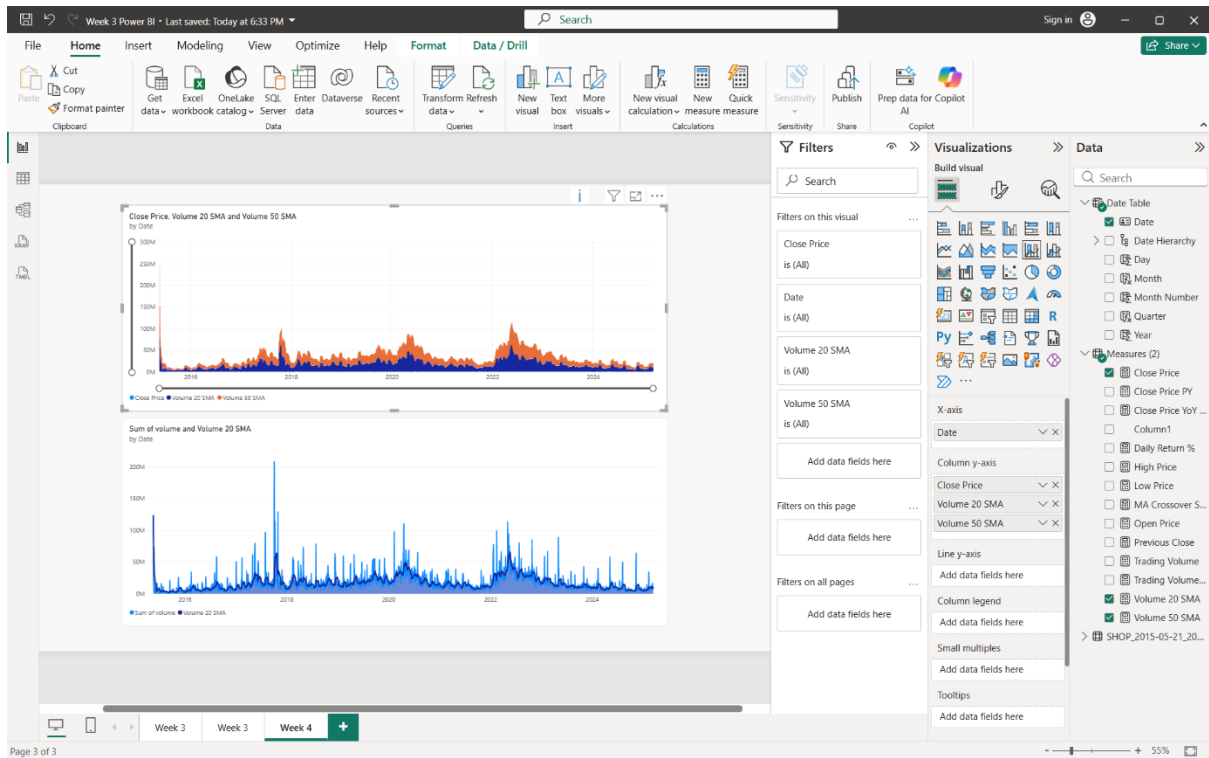
- **X-Axis:** Date Table[Date]
- **Line Y-Axis:** Close Price
- **Line Y-Axis:** Volume 20 SMA , **Line Y-Axis:** Volume 50 SMA

Set the X-axis type to **Continuous**.

To enable quick timeline navigation:

Format Visual → **X-axis** → **Zoom slider** → **On**

The chart displays the stock's closing price together with its 20-day and 50-day moving averages.



3. Trading Volume Visualization

Select a **Clustered Column Chart**.

Configure it as:

- **X-Axis:** Date Table[Date]
- **Y-Axis:** Total Volume
- **Y-Axis:** Trading Volume

The column chart displays daily trading volume, while the Volume 20 SMA shows the average volume trend.

Volume Spike Highlighting

A conditional color measure can be created to highlight unusually high volume.

Volume Color =

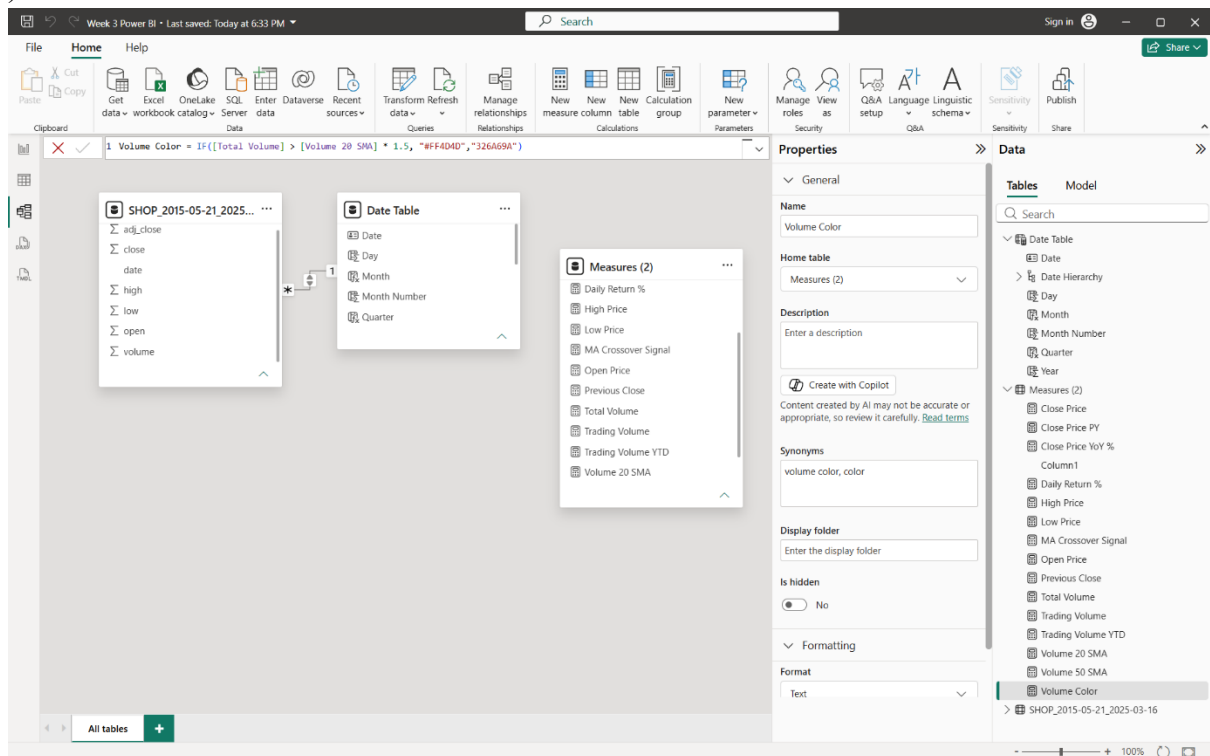
IF(

[Total Volume] > [Volume 20 SMA] * 1.5,

"#FF4D4D",

"#26A69A"

)

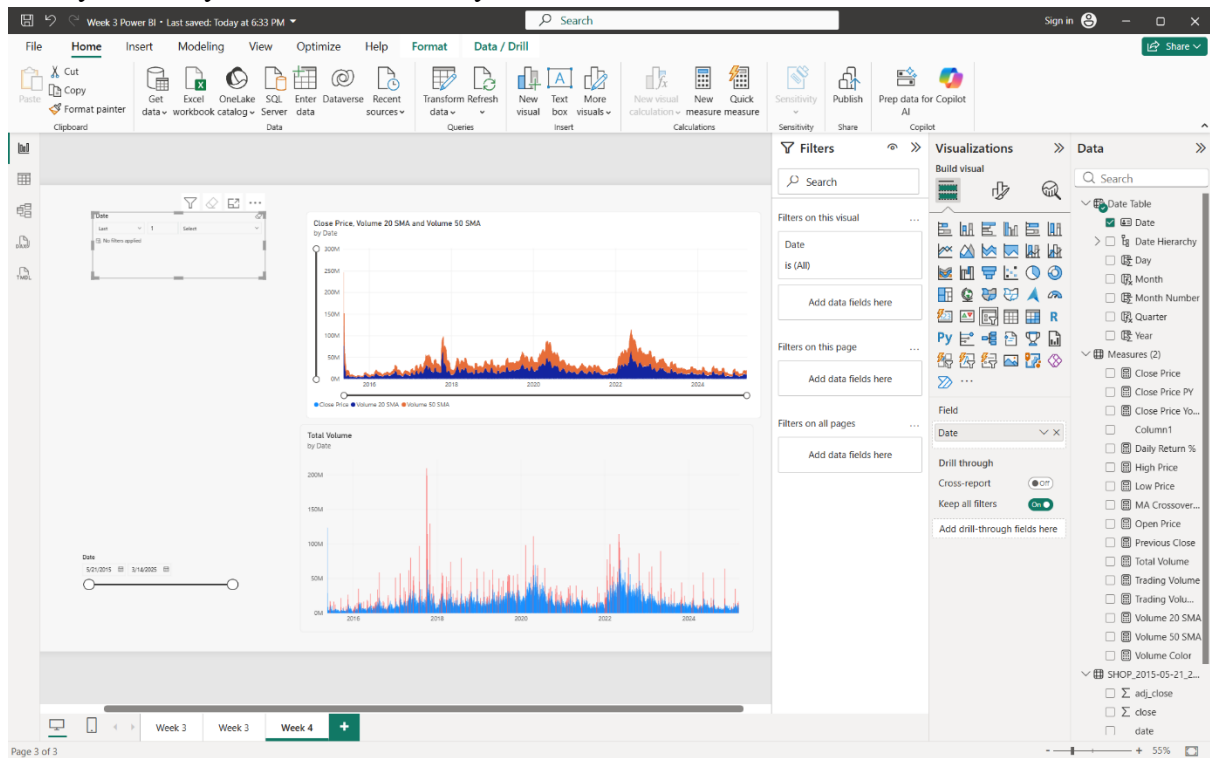


4. Timeline Slicer

Add a **Slicer** visual.

Drag: Date Table[Date] into the slicer.

Set the slicer type to **Between**. This allows the user to select a specific start date and end date and dynamically filter the stock analysis.



5. Metric Selector

To allow users to switch between different price metrics:

Go to:

Modeling → **New Parameter** → **Fields**

Create a parameter called:

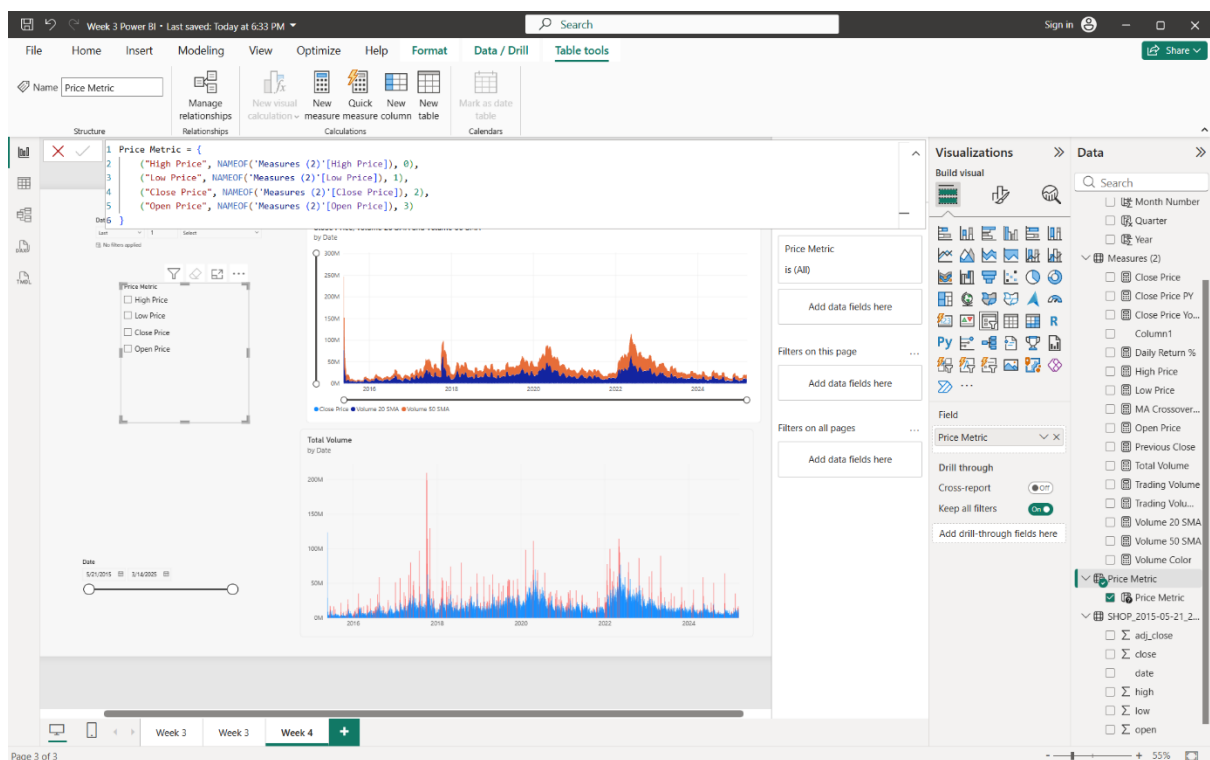
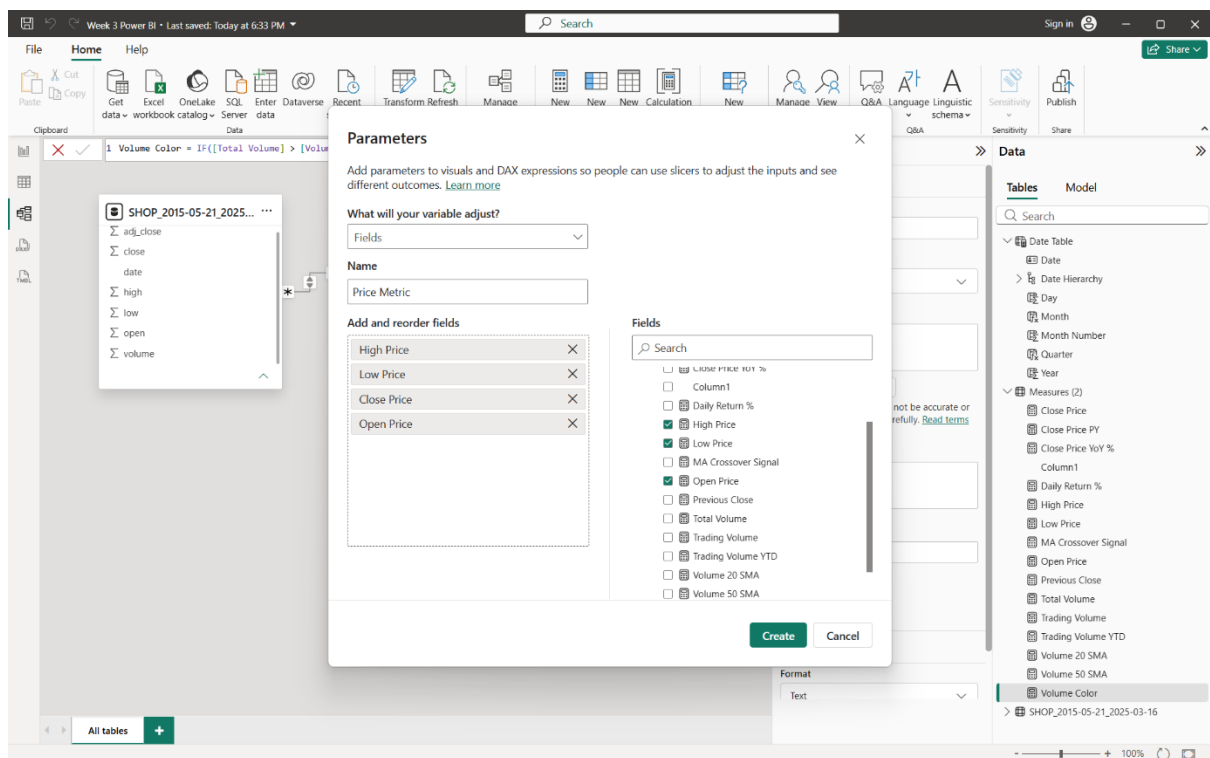
Price Metric

Add:

- Open Price
- High Price
- Low Price
- Close Price

Power BI creates a field parameter that can be placed in a slicer.

The user can then select the required price metric and dynamically change the visualization.



6. Custom Tooltip Page

Create a new report page and rename it:

Stock Tooltip

Go to:

Format → Page Information → Allow use as tooltip → On

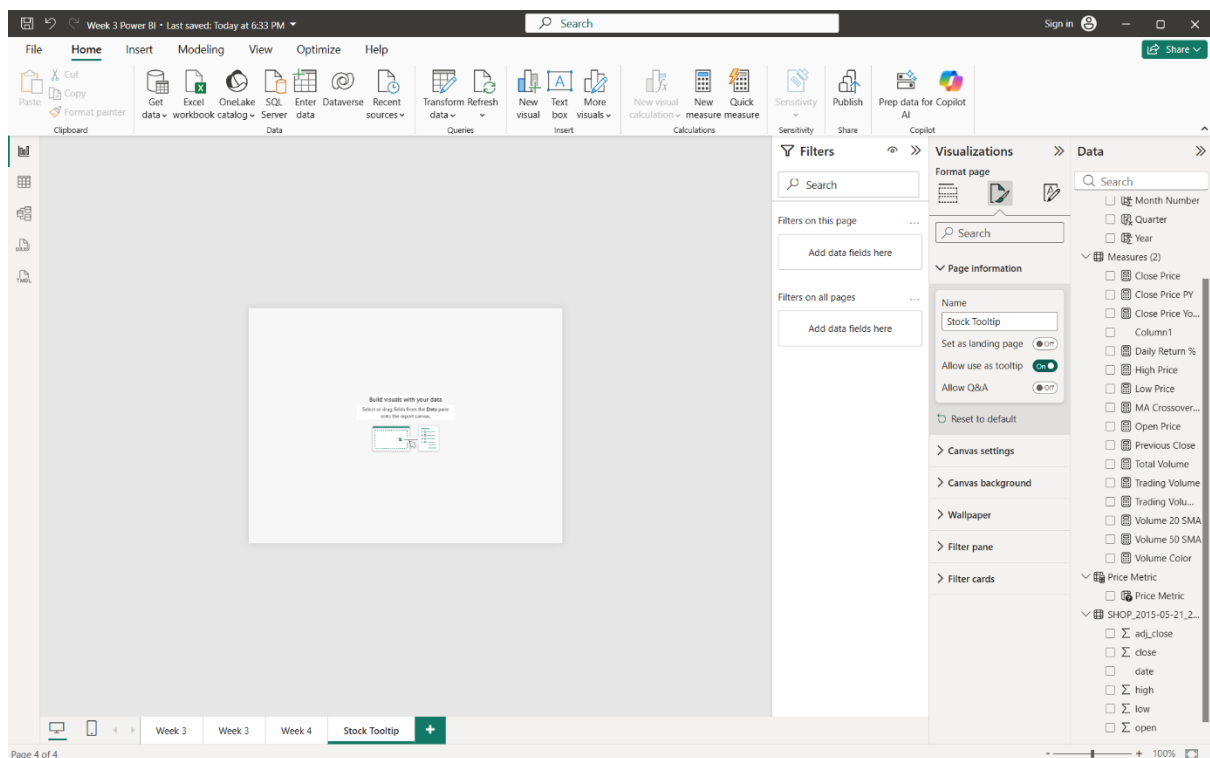
Then:

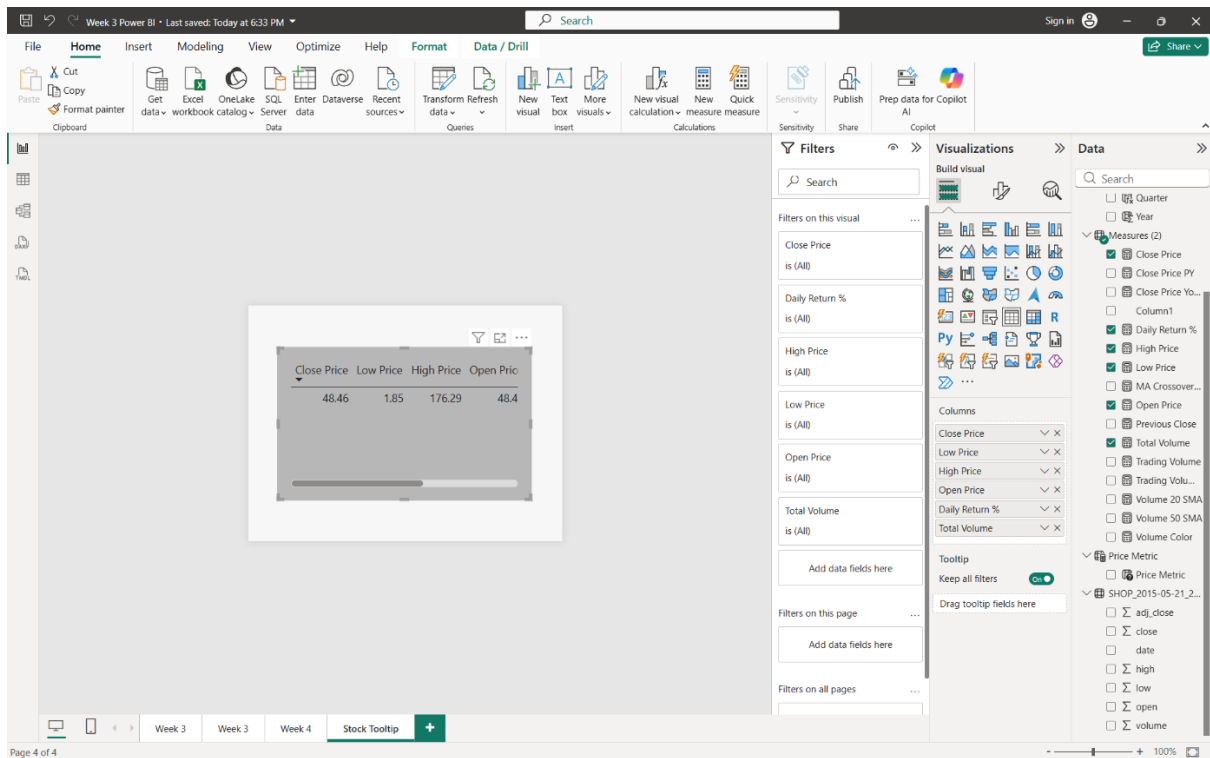
Canvas Settings → Type → Tooltip

Add Card or Table visuals containing:

- Open Price
- High Price
- Low Price
- Close Price
- Total Volume
- Daily Return %

This page provides detailed information when the user moves the mouse over a data point in the main price chart.





7. Connect the Tooltip to the Price Chart

Return to the main report page.

Select the **Price Trend Chart**, which is the **top chart containing Close Price, 20 SMA, and 50 SMA**.

Then go to:

Format Visual → General → Tooltips

Set:

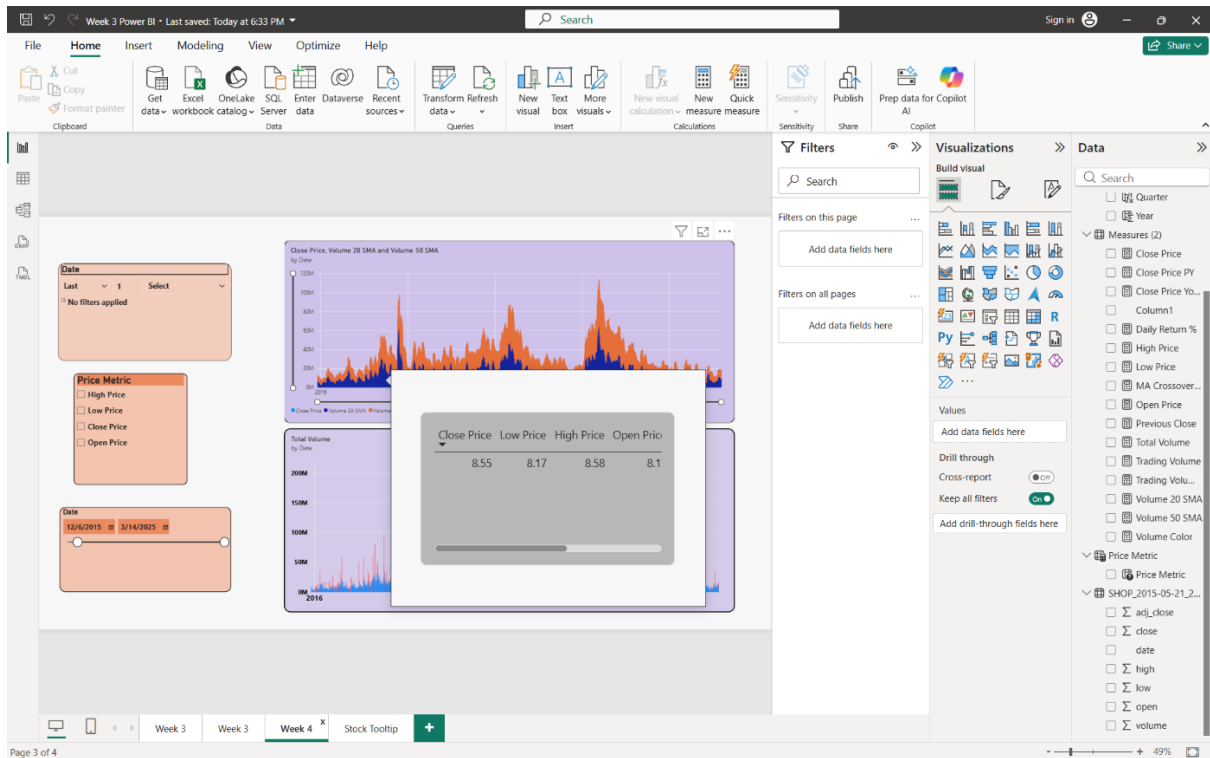
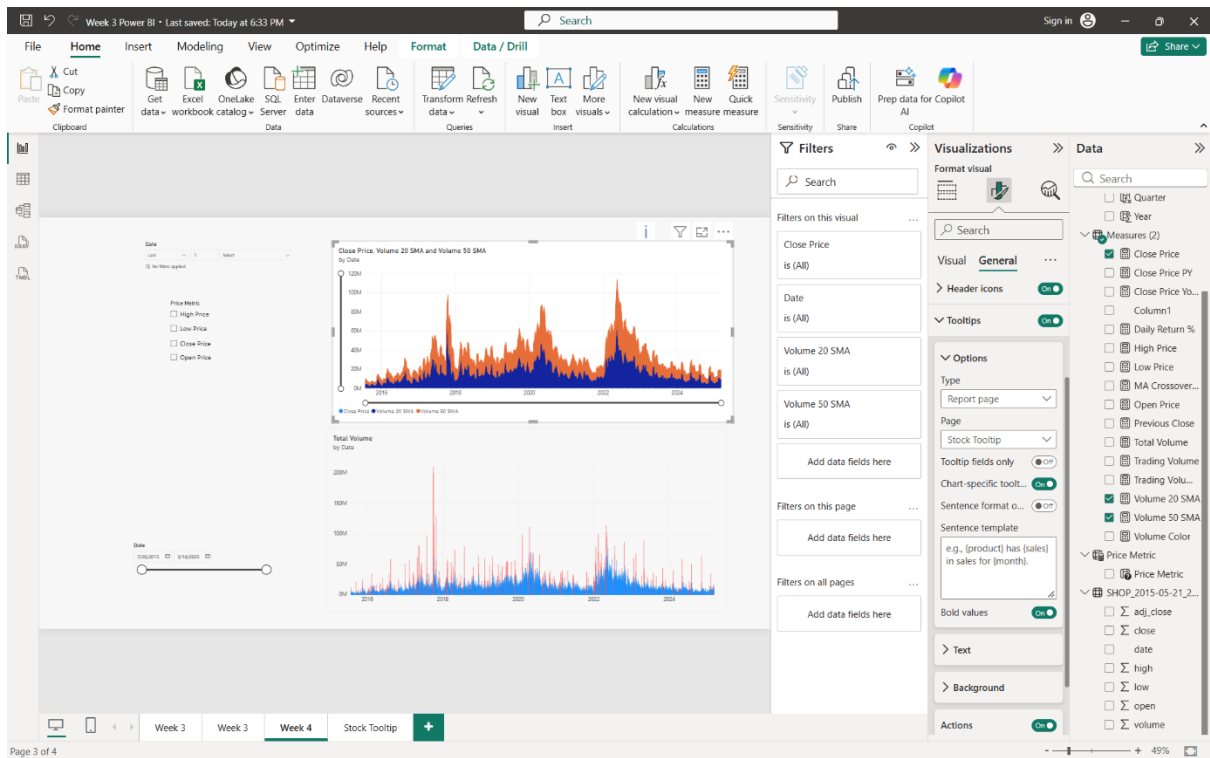
Type → Report page

Select:

Stock Tooltip

Now move the mouse pointer over a point on the price chart without clicking. This is called **hovering**.

The custom Stock Tooltip page will appear and display the detailed stock information for that date.



8. Interactive Dashboard

The final dashboard combines:

1. Stock Price Trend
2. 20-Day SMA
3. 50-Day SMA
4. Trading Volume
5. Volume 20 SMA
6. MA Crossover Signal
7. Date Range Slicer
8. Price Metric Selector
9. Custom Tooltip

These interactive elements allow users to explore stock price movements, trading volume, moving-average trends, and selected dates within a single Power BI report.



Conclusion:

The Shopify Stock Analysis dashboard uses DAX measures and Power BI visualizations to analyse historical stock prices and trading volume. The 20-day and 50-day SMAs help visualize short-term and long-term trends, while the volume analysis shows changes in trading activity. Interactive slicers, metric selection, and custom tooltips make the dashboard easier to explore and understand.